Question 18: Statistical Analysis of Age and Body Fat

18. Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result.

Question:

 Calculate the mean, median and standard deviation of age and %fat using Pandas.

 Draw the boxplots for age and %fat.

 Draw a scatter plot and a q-q plot based on these two variables

Aim: To calculate mean, median, and standard deviation and draw boxplots, a scatter plot, and a Q-Q plot.

Algorithm:

1. Store age and body fat data in a DataFrame.
2. Calculate mean, median, and standard deviation.
3. Draw boxplots for age and body fat.
4. Draw a scatter plot and a Q-Q plot.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

from scipy import stats

data = {

   "Age": [23, 25, 28, 30, 32, 35, 37, 40, 42,

        24, 27, 31, 34, 36, 39, 41, 45, 29],

   "Fat": [15, 18, 20, 22, 24, 25, 27, 29, 30,

        16, 19, 21, 23, 26, 28, 31, 33, 20]

}
```

```
df = pd.DataFrame(data)

print("Mean:\n", df.mean())

print("Median:\n", df.median())

print("Standard Deviation:\n", df.std())

df.boxplot()

plt.title("Boxplots of Age and Body Fat")

plt.show()

plt.scatter(df["Age"], df["Fat"])

plt.xlabel("Age")

plt.ylabel("Body Fat (%)")

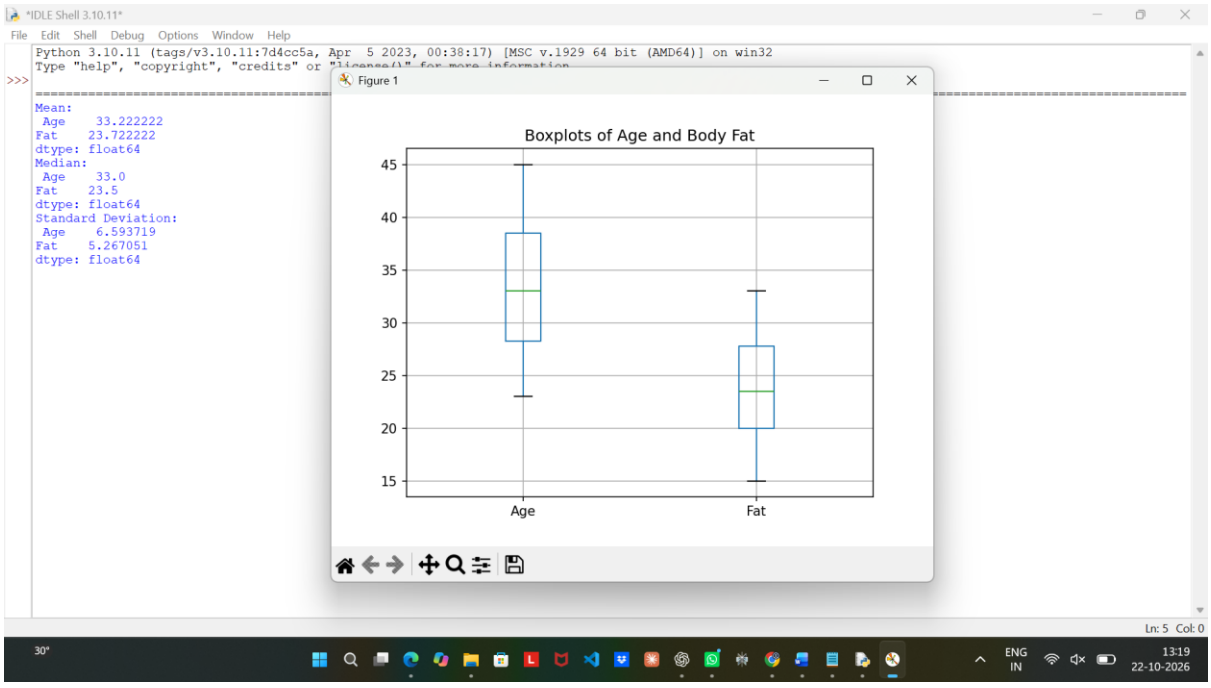
plt.title("Age vs Body Fat")

plt.show()

stats.probplot(df["Fat"], dist="norm", plot=plt)

plt.title("Q-Q Plot of Body Fat")

plt.show()
```



Note: The question does not include the original 18 observations, so the program uses sample data. Replace these values with the given hospital data.

Result: Statistical measures and the required plots were generated successfully.